Decentralizing Agriculture

A VERY NICE SUBTITLE

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1 Abstract

Here the abstract goes.

2 Introduction

3 The problems of industrial agriculture

Although the rise of the modern industrial structure of agricultural production has given many societies the ability to forestall famine and thrive in ways long thought impossible, many of its social and ecological repercussions have revealed themselves to be potentially dangerous in the best scenario and significant contributors to an existential crisis in the worst. Every step throughout the production process results in ecological problems of varying severity, from the clearing of the land to the delivery of the final product, each impacting the surrounding ecosystems and underlying soils in different ways. It is a production model that forces the whole population into the dependence of a few producers, who have increasingly incorporated more and more monoculture practices, thus making the food supply more vulnerable to collapse in the case of an uncontrolled pest. The following sections will explore some of the ecological and social problems related to industrial agriculture. [TODO: EXPAND THIS INTRO ONCE YOU'RE DONE WITH THE SUBSECTIONS]

3.1 Land Use

The prevailing paradigm of industrial agricultural production requires the use of vast swathes of land, along with a considerable amount of water, energy and other resources, which inevitably ends up homogenizing the local flora and destroying the habitat of the local fauna, leading to an overall loss of biodiversity. The agricultural expansion that the current system entails is indeed one of the main drivers of global biodiversity loss (IPBES, 2019, p. 412; European Commission, 2024, p. 199), and is the most widespread form of land-use change, with cropping and animal husbandry covering over one third of the terrestrial land surface (IPBES, 2019, p. XVI).

[TODO: see if you can find a quantitative biodiversity loss metric]

This geographical expansion of the agricultural sector has been a constant for decades, having grown by 78 million ha (780,000 km^2) between 2001 and 2023, mostly driven by an increase in permanent crops such as cocoa, oil palm or coffee (FAOSTAT, 2025). All of these are used intensively for monoculture, as are feedstock crops which by themselves account for one third of the total crops produced globally (Battersby et al., 2024, p. 22); this particular form of agriculture, which consists in the the use of a single crop for large productions, is another potent driver of biodiversity and habitat loss which will be explored in a subsequent section.

The extension of anthropogenic drivers of change has led to as little as 13% of the ocean and 23% of the land falling under the classification of 'wilderness'. These few remaining areas that lie still outside of humanity's sphere of influence tend to be remote or unproductive, like tundras or oceanic gyres, the most hospitable biomes having been almost or completely modified by humans in most regions (IPBES, 2019, p. 206). With the transformation for human convenience of the landscapes of the world that are the most accommodating to life, displaced wild populations are left without a habitat.

The impact of this intrusion into the wilderness has also far-reaching consequences as well. According to the FAO's State of Food and Agriculture report (FAO, 2025, p. 4), agriculture currently drives 90 percent of global deforestation, with cropland expansion accounting for almost half of the total deforested area.

This leads to changes in both global and local climates through two main mechanisms: the release of stored carbon produces a robust warming signal worldwide, while the removal of forest cover changes surface albedo, evapotranspiration and canopy roughness, which produces a global warming effect for deforestation in the tropics, and a cooling effect at higher latitudes. At the local scale, forest loss almost always raises surface temperatures, with typical increases between 0.2 °C - 2 °C, depending on latitude (Lawrence et al., 2022).

3.2 Soil degradation

4 Background

Past attempts, Netherlands case study

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